Criterion A: Planning

1. Defining the problem

My client, Ryan Zhang, an LOHS TV gameday commentator, wants an easy way to record the stats for every player on the Lillian Osborne basketball team. To record the stats of every player in high school basketball, their stats have to be typed into a spreadsheet which is inefficient and time consuming. Problems include inaccurate stat tracking and limited stat recording. I held an interview with Ryan to discuss how a simple way to record stats can be made. The interview allowed Ryan to describe how he expects the program to function. Ryan mentioned that he would like a program that is able to do calculations rather than having to do them manually.

My client suggested that I made a Python program to allow users to interact with the stats database. A solution that I proposed is to make a database to store data while displaying the stats in a nice table. The program will be user friendly and easy to input information. It will be easy to input new information as well as edit the existing information.

I will hold a second interview with Ryan so that he can share his opinion on how the program should be used and his experience with the program. Each interview with my client is documented in the appendix.

2. Rationale for proposed product

The idea of the program is to store and display information, as well as calculate averages of the data inputted. Users will be able to add data, delete data and view existing data in a nice and organized way. The program will be a user-friendly way for Ryan to view stats of players.

In order to create an interactive database, I believe that SQL and Python is an effective way to achieve this. I will be using the “sqlite 3” library in Python, as well as “tabulate” for displaying tables, and “numpy” for calculations.

SQL is a language that I am familiar with and have little problem managing a database with SQL. The language allows easy interaction with databases through python code.

3. Success criteria

- Program will display a menu
- Program will display data in a nice table
- Program will ask for the name of the player to be added
- Program will ask for a value to be inputted for a stat
- Program will be able to display the names of every team member
- Program will be able to calculate averages
- Program will restart without the user having to restart it manually

Word count: 419